BIS3216 DATA MINING AND KNOWLEDGE DISCOVERY

ASSIGNMENT 1

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1.0 Introduction

Data preprocessing is an essential step in any data science project. It involves cleaning, organizing and transforming raw, messy data into a format that is usable to be analysed effectively. When done correctly, preprocessing provides machine-learning models the ability to work with accurate and consistent data for more reliable results and clearer insights. When the input is of poor quality, the results will be inaccurate or misleading and weaken the overall performance. Priestley et al. (2023) explained that the outputs produced by models are directly influenced by the quality of data used to train them. Likewise, Similarly, Foidl et Al. (2023) emphasized that the stability and reproducibility of ML workflows depend heavily on the management of the preprocessing and the pipeline stages. As datasets grow larger and more diverse, data preprocessing remains an important step towards building reliable and interpretable analytic systems.

1.1 Problem Context

Real-world data rarely come in perfect condition. Data sets might have holes, inconsistencies, or junk that can distort patterns and decrease the accuracy of machine-learning models. Unreliable data may have missing values, incorrectly formatted values, or duplicate/incorrect entries. According to Puchakayala et al. (2022), the performance degradation of a model can be prevented by sustaining the quality of data throughout the ML pipeline. Macht and Brogly (2022) also refer that the bias and fairness of algorithms are adversely affected by poor data quality, which results in risk across AI applications other than medicine. Priestley et al. (2023) explained that clearly defining data-quality requirements early in the workflow helps avoid reliability issues later on. In the same way, Foidl et al. (2023) stated that pipeline quality helps to improve the accuracy and stability of models over time. To sum up all, through accurate preprocessing and validation of data, the robustness, fairness, and overall reliability of ML systems can be ensured.

2.0 Method by Data Type

2.1 Structured or Tabular Data

Tabular data is arranged in rows and columns. It is commonly in databases, spreadsheets and CSV. files. It helps in prediction, classification and anomaly detection task. However, these datasets often contain missing values, unscaled features, categorical variables and outliers which can affect model performance if not handled properly.

Missing-Value Handling (Imputation)

When some of the data entry values are blank due to human, sensor or system errors, we have missing values. According to Shadbahr et al. (2023), poor handling of missing data results in inaccurate predictions and bias. Approaches like mean, median, or mode are fast and easy but might cause distortion in the relationships. More advanced methods such as k-Nearest Neighbour and MICE improve completeness by estimating missing values based on feature relationships.

Scaling / Normalization

Feature scaling adjusts numerical ranges to ensure all variables contribute equally to model training. Pinheiro et al. (2025) assessed scaling effects on regression and classification models and found that. Normalization methods, such as Min-Max and Z-score, make the algorithm more stable. These methods reduce magnitude bias and speed up convergence, especially in distance- or gradient-based models.

Encoding

Before performing analysis, categorical attributes must be converted into numerical values. Matteucci et al. (2023) benchmarked several encoding techniques and found that *one-hot* encoding performs best for small outputs, while *target* and *binary* encoding are more efficient for high-cardinality features. Encoding reduces inconsistencies and helps algorithms process non-numeric information effectively.

Outlier Detection and Treatment

Outliers are extreme values that deviate from normal patterns. Most often caused by data entry errors or rare events. Badhan and Ganpati (2024) reviewed common detection methods such as Interquartile Range (IQR), Z-score, and Isolation Forest. These methods identify anomalies statistically or through tree-based separation. Winsorization and trimming can be applied to limit the influence of extreme values, enhancing model robustness.

Duplicate Detection

Duplicate records occur when the same entry is stored multiple times mainly due to data integration or human errors. Azeroual et al. (2022) proposed a framework based on record linkage that uses matching algorithms and hashing for the effective detection of duplicates. The removal of redundant records ensures consistency in data and prevents biased results from statistical analyses.

Class Imbalance Handling

Class imbalance occurs when some categories have significantly more entries compared to others, which might lead to biased predictions. Chen et al. (2024) discussed techniques that can be performed for balancing class distributions, including Synthetic Minority Over-sampling Technique (SMOTE), undersampling and cost-sensitive learning. These methods enhance the fairness and improve generalization across categories.

2.2 Textual Data

Textual data is defined as information that is shown in written or printed words which is often unstructured or semi-structured. Data preprocessing is addressed to transforms this raw text into structured, numerical format. (Caterino, n.d.)

Noisy-text cleaning

When data are first gathered from any sources, it can have irrelevant or unwanted information presents in the datasets, these 'noise' can be removed by applying a set of rules to identify meaningful patterns and insights. (HaCohen-Kerner et al., 2020) The common techniques use include removing punctuation that does not serve any meaning such as are commas, quotes and others. (Kathuria et al., 2020)

Tokenization

Tokenization is the process of creating basic structure from cleaned text. This functions by segmenting a string of text into a list of individual units **(token)**. These tokens became manageable that can use for analysis. However, overdoing tokenization can cause significant issue that lose content and meaning in natural language processing. (Kathuria et al., 2020)

Lemmatization

Lemmatization is the technique of reducing words to their root form that still contain its meaning. It involves using dictionary as vocabulary to analyse words to remove inflectional endings and to return the base form of a word, which is known as a lemma. This technique significantly decreases the volume of vocabulary to improve accurate data mining while reduce space. One common technique is stemming which is the process of reducing words to their root or base form without removing it's meaning to improve text analysis such as "presenting", "presented" reduce into "present". Another is removing stop words which are words that occur frequently without any actual meaning such as "and", "are", "this" and more. (Kathuria et al., 2020)

Embedding

Embedding is the technique of translating words into numerical format **(vectors)** that machines can understand. This numerical representation is learned from data that captures the object's semantic meaning and its relationships to other objects, enable machine to perform tasks in similarity search and classification. (Amazon Web Services, 2025) A common technique is Natural Language Processing (NLP), where machine can understand and communicate with human language. (Stryker & Holdsworth, 2024) For example, vectors with similar meaning such as "dog" and "puppy" will be mathematically close, while for vector like "car" and "water" will be distant.

Transformer-based methods

Transformer-based methods is an advanced deep learning architecture that processes text by understanding the nuances of language. Its core technique is self-attention, which is all tokens in a sentence at once. By weighting the importance of each word in a relation to all other words to understand the true meaning. A common concept includes **contextual encoding** which generating contextual meaning such as differentiate the meaning of “river bank” and “bank account”. (Manoj Prabaharan, 2025)

2.3 Image Data

Image data provide visual information that stored at discrete locations called pixels. However, it often affects by quality issues such as sensor noise, low resolution and artifacts that reduce visual clarity and affect analysis accuracy. Thus, preprocessing technique like image denoising, contrast enhancement, resizing, normalization, augmentation, correction of low-resolution is used to fix them.

Image Denoising

Liu et al. (2022) highlighted denoising address noise caused by environmental and sensor limitations. It will solve the issues such as inconsistent resolutions, aspect ratios, and object scales that occur when images are collected from different sources. Traditional methods like BM3D and total variation rely on mathematical priors. Deep learning models such as DnCNN and FFDNet learn mappings between noise and clean images. To overcome over-smoothing and loss of detail problem, the Flow-Based Denoising Network uses normalizing flows to disentangle clean image information from noise, producing sharper, more realistic image without assuming specific noise type.

Contrast Enhancement

Contrast enhancement is the process of adjust the brightness and colour differences of an image to improve the visibility This improves medical images by highlighting important texture changes and correcting illumination inconsistencies. According to Heller et al. (2021), this technique used CT images in the KiTS19 study involve methods such as Hounsfield Unit clipping, percentile-based scaling, and normalization to zero mean and unit variance. These techniques help to enhance local contrast and correct intensity imbalance. As a result, contrast enhancement promotes consistent feature representation by improving segmentation accuracy and model robustness across diverse imaging conditions.

Resizing

Image resizing changes the dimension by adjust the number of pixels of the image. This method uses to solve problems such as inconsistent image dimensions, aspect ratios, and object scales. According to Talebi and Milanfar (2021), a CNN-based learned resizer can be jointly trained with vision models to optimize resizing for recognition tasks. The resizer will perform downscaling or upscaling to a fixed resolution before feeding them into models such as ResNet or Inception. This ensures consistent input for accurate recognition.

Image Normalization

Image normalization will modify pixel value to enable best comparison across various data gathering techniques and texture instance. This process is important for imaging modalities that do not correspond to absolute physical quantities. According to Huang et al. (2023), the common normalization techniques had included centering, scaling, standardizing, and whitening. They reduce the difference in magnitude between image features and improve model convergence. They also mention Batch Normalization as an important technique that standardizes activations within mini batches to stabilize training. Besides that, layer normalization and instance normalization extend this to individual samples and images.

Image Augmentation

Based to Shorten and Khoshgoftaar (2019), image augmentation increases the size and diversity of image datasets by using label-preserving transformations. One of the transformations is geometric operations. It performs operation such as rotation, flipping, cropping, and scaling, colour and intensity adjustments (brightness, contrast, and noise addition). It improves model generalization and reduces overfitting, especially training data are limited. Besides that, advanced augmentation methods such as neural style transfer, generative adversarial networks and feature space augmentation will create realistic and diverse variations of training images.

Correction for low resolution

Low resolution correction used to enhance image quality and resolution. This process produces high-resolution images from low-resolution inputs by recovering lost details and sharpness. Based on study from Messina et al. (2002) proposed an adaptive back-projection algorithm that improves classical iterative back-projection methods through local uncertainty estimation and auto-iterative correction. This approach will improve pixel reconstruction, reduce computational time, and preserve important spatial details that result in sharper and more accurate high-resolution images suitable for real-time processing.

2.4 Audio Data

Noise Reduction

Raw audio signals can have a number of flaws like background noise, poor and inconsistent levels of volume, clipped waveforms and too much silence. These problems are barriers to proper pattern recognition and model performance. In addition, it was highlighted in the study that this quality deterioration is common in practical sound recording, especially in a setting with crowd, machine, or wind (Abayomi-Alli et al., 2022). Subsequently Nogales et al. showed that unfiltered noise has the capacity to mask essential sound characteristics, whereas untrimmed silence fools sequence-based classifiers, so prior preprocessing is thus critical to noise classification or detection activities. (Nogales et al., 2023)

Silence Trimming

A good preprocessing stage starts with the cleaning of the audio stream. This involves denoising filters to keep down background noise and silence trimming to eliminate dead periods that add padding and delay. According to Chu et al. (2023), it is important to normalize the amplitude of signals and trim silence before converting signals. Likewise, environmental distractions like traffic or wind noise were removed by convolutional variational autoencoders and greatly enhanced intelligibility. These guiding stages narrow down crude input into a smaller and informative size that can be used in training of models. (Nogales et al., 2023)

Sampling-Rate Conversion

Sampling-rate conversion A conversion of audio signals between sampling rates (usually between two related sampling rates), most commonly to standardize input between datasets. Unsteady sample rates are capable of distorting the time resolution and influencing the consistency of the features that are extracted in sound. Most preprocessing pipelines do not explicitly address the experimental details of resampling, however most of these pipelines use resampled audio as a starting point before extracting MFCC or STFT. As an example, Abayomi-Alli et al. (2022) mention that irregular volume and waveform clipping that are typically accompanied by resolution problems make it difficult to identify patterns. Switching to fixed sampling rate makes extracted spectrograms (e.g., log-Mel or FFT) have a uniform temporal scale- a necessary since transformer-based audio models usually require it (Zaman et al., 2025). The step helps him to avoid aliasing, enhance feature alignment, and facilitates cross-dataset model training.

Segmentation

Segmentation is a preprocessing method that pre-cuts long or continuous audio into small and manageable frames or chunks to do the analysis. This is necessary in the case of real-world recordings which are not consistently structured or labelled. Models can utilize the concept of lacking localized sound events and enhancing recognition accuracy through the ability to slice audio into overlapping or fixed-size windows. Though not the focus of all research, Chu et al. (2023) used segmentation when creating MFCC spectrograms using short audio slices, which enabled the model to differentiate between sound events of interest. Moreover, segmentation allows more efficient data augmentation, since they can be changed, or resampled one short frame at a time. Effective

segmentation enhances learning effectiveness and makes sure that models are not overloaded with useless silence or noisy areas.

Feature Extraction

A significant stage of preprocessing is feature extraction, which transforms raw audio waveforms into numerical data, including MFCCs and STFTs, which describe important spectral and temporal properties. It helps in applications such as speech recognition, tagging music, and classification of sounds in the environment. A study has shown that a comparative analysis based on MFCC and STFT features on the UrbanSound8K dataset with artificial neural networks was successful at over 90 percent. (Gourisaria et al. , 2024)). Besides, demonstrated that the CNN-based MFCC models that are further augmented with data enhance performance.(Chu et al., 2023) MFCCs are also background noise sensitive (Nogales et al., 2023) and STFTs are computationally expensive. Study also highlighted that transformer models work optimally when preprocessing is clean and consistent, necessitating a trade-off between scalability and data variability (Zaman et al. ,2025; Abayomi-Alli et al., 2022).

2.5 Video Data

Frame Sampling

Frame sampling selects a subset of frames from a video to reduce redundancy, capture important information and decrease data size. Kandhare and Gisselbrecht (2024) emphasize that the technique extracts key frames that preserve semantic coverage without storing every frame. Their study demonstrates several sampling strategies including interval selection, pixel and histogram-based change detection, structural similarity, semantic embedding comparison and shot-boundary detection that can achieve recall scores comparable to storing all frame and lowering the number of frames indexed and retrieved.

Temporal Segmentation

Temporal segmentation divides the continuous video based on time or content as a scene. It handles video-related quality issues from time continuity, scene transitions, or content inconsistency across frames. Xin et al. (2022) introduced Scene Separation and Data Selection algorithm that detects changes between frames by using an improved difference hash method to measure visual similarity. Frames with small difference are grouped into the same scene to allow the algorithm preserve essential temporal information while reducing redundancy.

Motion Stabilization

Motion stabilization will remove unwanted camera shake and produce smooth video. This technique removes the unintended camera movements by estimating and correcting motion trajectories. According to Lee et al. (2021), the Deep3D Stabilizer integrates depth estimation and 3D camera motion modelling using convolutional neural networks (CNNs). This method reconstructs scene geometry and smooths camera trajectories through a 3D optimization framework, enabling real-time stability manipulation.

Corrupted-frame removal

Corrupted-frame removal will identify and delete individual frames that are unreadable or contain error in a video due to transmission errors, compression faults, or storage corruption. It removes the frames to prevents artifacts that reduce overall video quality. Besides that, it ensures that subsequent recovery and analysis methods operate on clean and consistent data. Liu et al. (2023) proposed a bitstream-corrupted video recovery method (BSCVR) that automatically detects and reconstructs corrupted frames using a transformer-based video inpainting approach. It detects corrupted regions within frames and reconstructs them using a transformer-based video inpainting network that combines spatial and temporal information from neighbouring frames.

2.6 Time-series/ Sensor/ IoT Data

Resampling

IoT sensor data will be prone to gaps in timestamps, sampling anomalies, signal drift and noise. According to a study by Stefanou et al. (2025), motion data collected by wearable devices like accelerators tends to be variable because of sensor misplacement or motion artifact by the user influencing the quality of the signal. Temporal alignment and predictive modelling have become more challenging due to signal baseline drift and irregular sampling intervals. (Stefanou et al., 2025) To standardize these irregularities, resampling methods, including time-based aggregation, up sampling or interpolation-based smoothing, are used to rebuild constant time intervals. These problems should be implemented in the pre-processing so that meaningful analysis of the AI can be done.

Smoothing

The time-series and IoT data are typically fraught with missing values, signal drift, non-stationarity and outliers. As a solution to these, already developed preprocessing methods, like interpolation (linear, spline, polynomial) are introduced to determine the missing values. To reduce noise, there are smoothing algorithms such as the exponential moving average. Furthermore, Box-Cox transformations or seasonal decompositions (e.g., STL) are used to ensure that the variance and trends are stabilized (Usmani et al., 2024; Tawakuli et al., 2025). Z-scores, clustering, or autoencoders based on deep learning can be used to identify and filter outliers (Wang et al., 2025).

Seasonal-trend decomposition

After cleaning, the raw sensor signals are sliced and converted into informative learning representations. Following, Fixed/adaptive/windowing provides the opportunity to slice continuous data streams, whereas such techniques as feature engineering as PCA or FFT can be used to capture both patterns based on time/frequency (Tawakuli et al., 2025). Moreover, multi-sensor streams are to be fused by matching time stamps which is important to assure time correctness in the IoT application. Besides, seasonal-trend decomposition techniques like the STL are also used to distinguish between long-term trends and short-term variations to enhance the interpretability of the model and reduce residual noise before training.

Anomaly Detection

Good preprocessing also entails the detection and removal of anomalies, outliers, unwonted spikes, or erroneous sensor measurements, which may bias learning. Univariate or low-dimensional streams

are often analysed with statistical tools like z-score analysis, filters based on IQR and isolation forests. In deep learning, autoencoders, recurrent neural networks and GAN-based detectors are used as suitable deep learning techniques to learn normal behaviour and identify deviation in real time. A preprocessing pipeline can perform the techniques at the preprocessing stage to clean training data and minimize false alarms in further models particularly in systems that are safety-conscious, such as health monitoring or industry error detection. (Wang et al., 2025).

Sensor Drift Correction

In order to minimize human intervention automated preprocessing pipelines like Preptimize utilize meta-learning to suggest the best strategies to each dataset (Usmani et al., 2024). Modular ETL frameworks running on edge devices (implemented as containerized environments, e.g., Docker/Kubernetes) are used on the hardware level to support low-latency, scalable preprocessing (Urblik et al., 2023). Besides, Mohammed et al. have suggested a deep reinforcement learning (DRL) agent that cleans sensor data in real-time with high accuracy at a low computational cost. (Mohammed et al., 2023)

2.7 Graph/Network Data

Graph data represent information structured through entities (nodes) and their relationships (edges), where nodes serve as data points and edges indicate their connection. (Enzo, 2025)

Node/edge consistency checks

Node/edge consistency checks serve as an error detection process that identify data that violate the established rules for what the data should look like and what it should mean. Error detection techniques check for structural inconsistencies, where relationships point to entities that do not exist to ensure the graph respects simple integrity constraints. The technique also handle redundancy, identifying repeated facts or relations in the graph that can be considered errors in certain contexts. (Xue & Zou, 2022)

Handling Missing Attributes

Handling missing attributes use techniques to address gaps in a dataset by using the three approach which are deletion, single imputation and advanced imputation. (Heymans & Twisk, 2022) The common technique uses include Neighbour-based imputation which works by filling a node's value with the mean, median, or mode of its immediate neighbours. Another is State-of-the-art GNN imputation which views this as a prediction problem, where a model is trained on the portions of the graph that are known to learn the relationship between structure and attributes, enabling the model to predict the missing values. (Xia et al., 2025) This addresses the incompleteness of data based on the structure of the graph and creates much richer and complete dataset, allowing data mining model to find more accurate patterns.

Graph Sampling

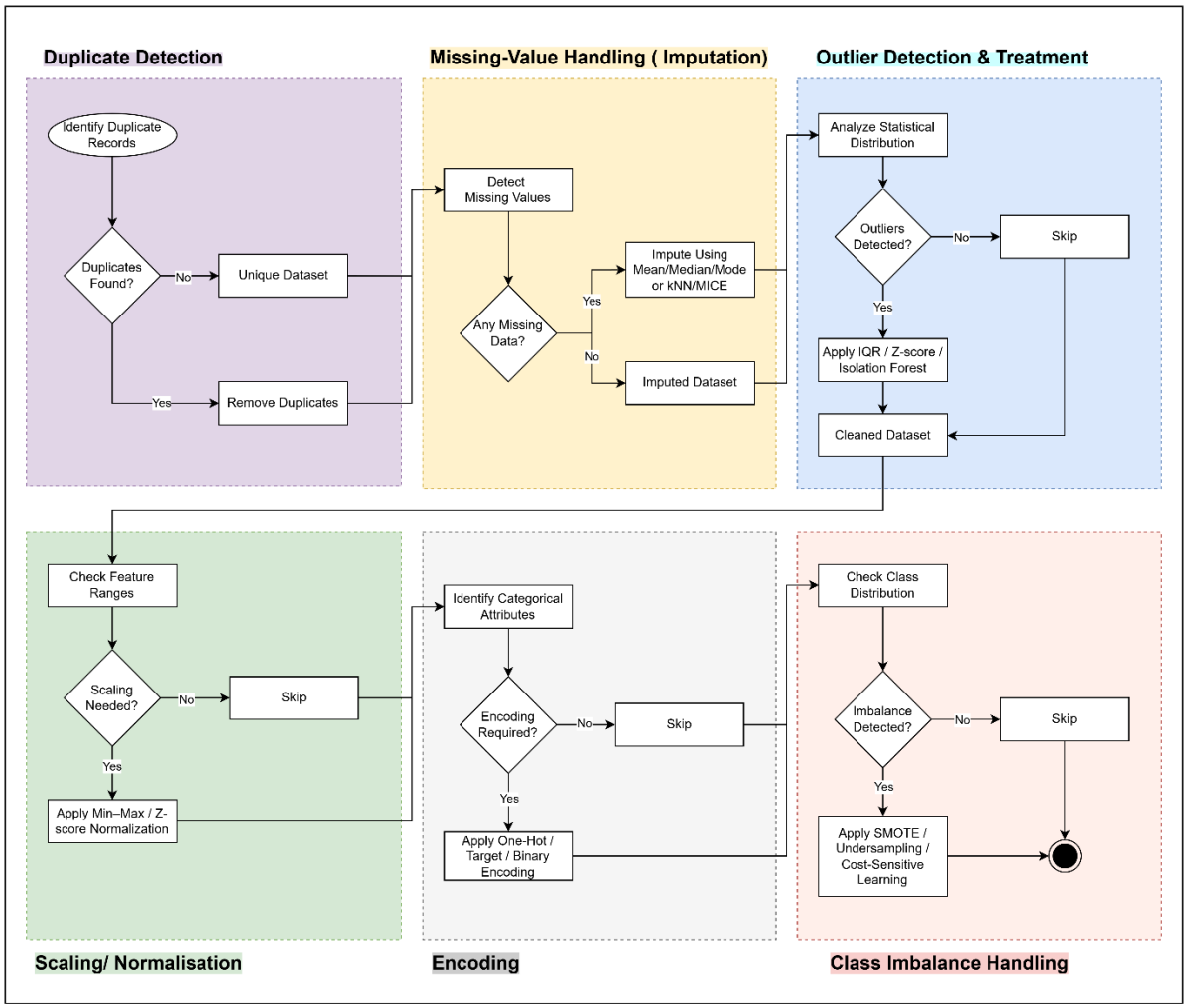
Graph sampling is a technique that extract smaller subgraph from large graph while maintaining its key meaning. (Jiao, 2024) According to Hu & Lau (2013), the common techniques include edge sampling randomly selects a subset of edges, together with the nodes connected by these edges. Moreover, random walk sampling process by start exploring from a random node and travers to its

neighbours for a certain number of steps, collecting visited nodes and edges which is effective in preserves the local structures. These techniques address the scalability problem of large networks by drawing subgraphs that are representative.

2.8 Process Diagram

Figure 1

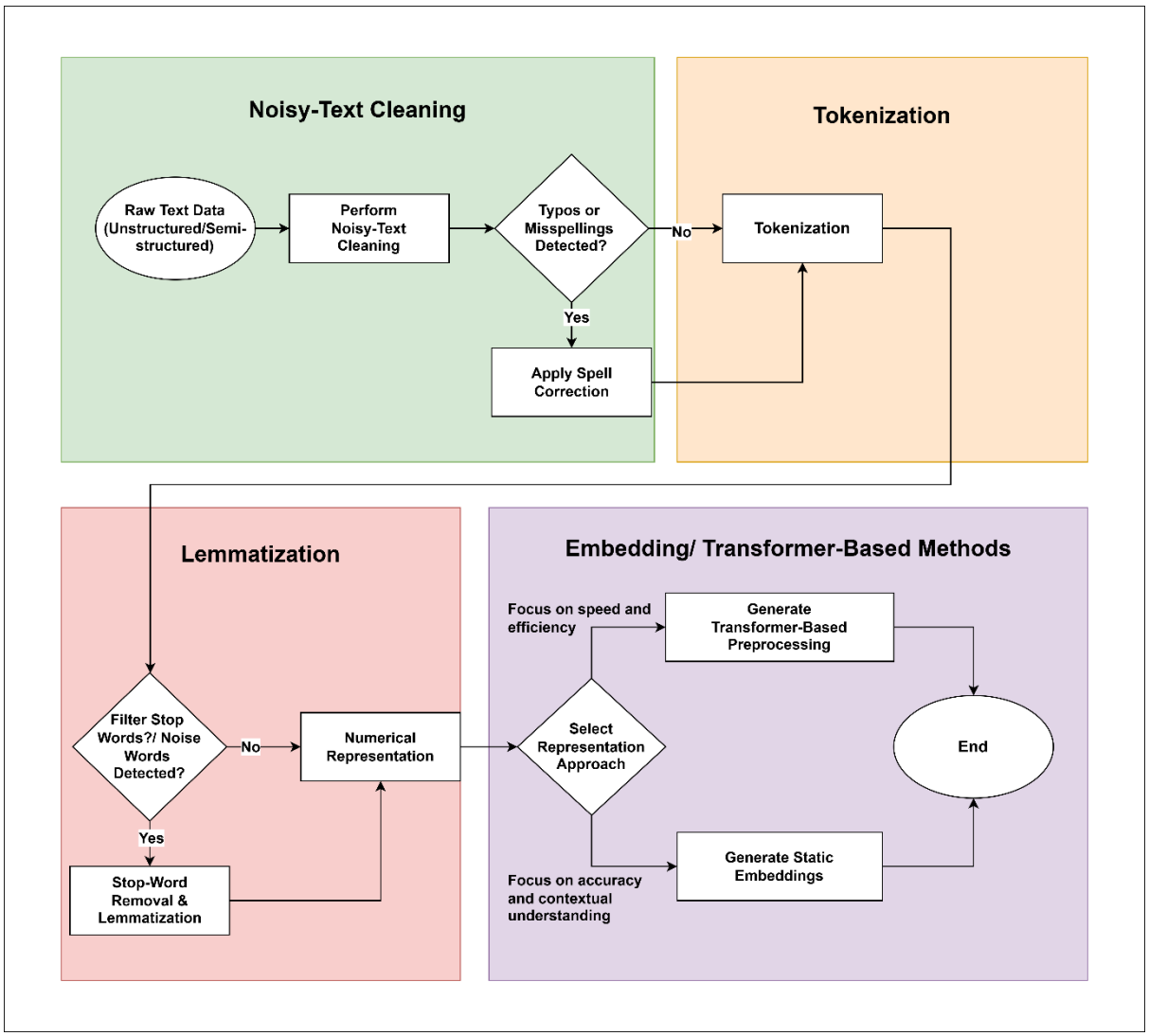
*Flow Chart for **Structured Textual Data** Preprocessing Techniques*



Note. This sequence follows standard data-cleaning and preparation practices supported by recent research (Pinheiro et al., 2025; Shadbahr et al., 2023).

Figure 2

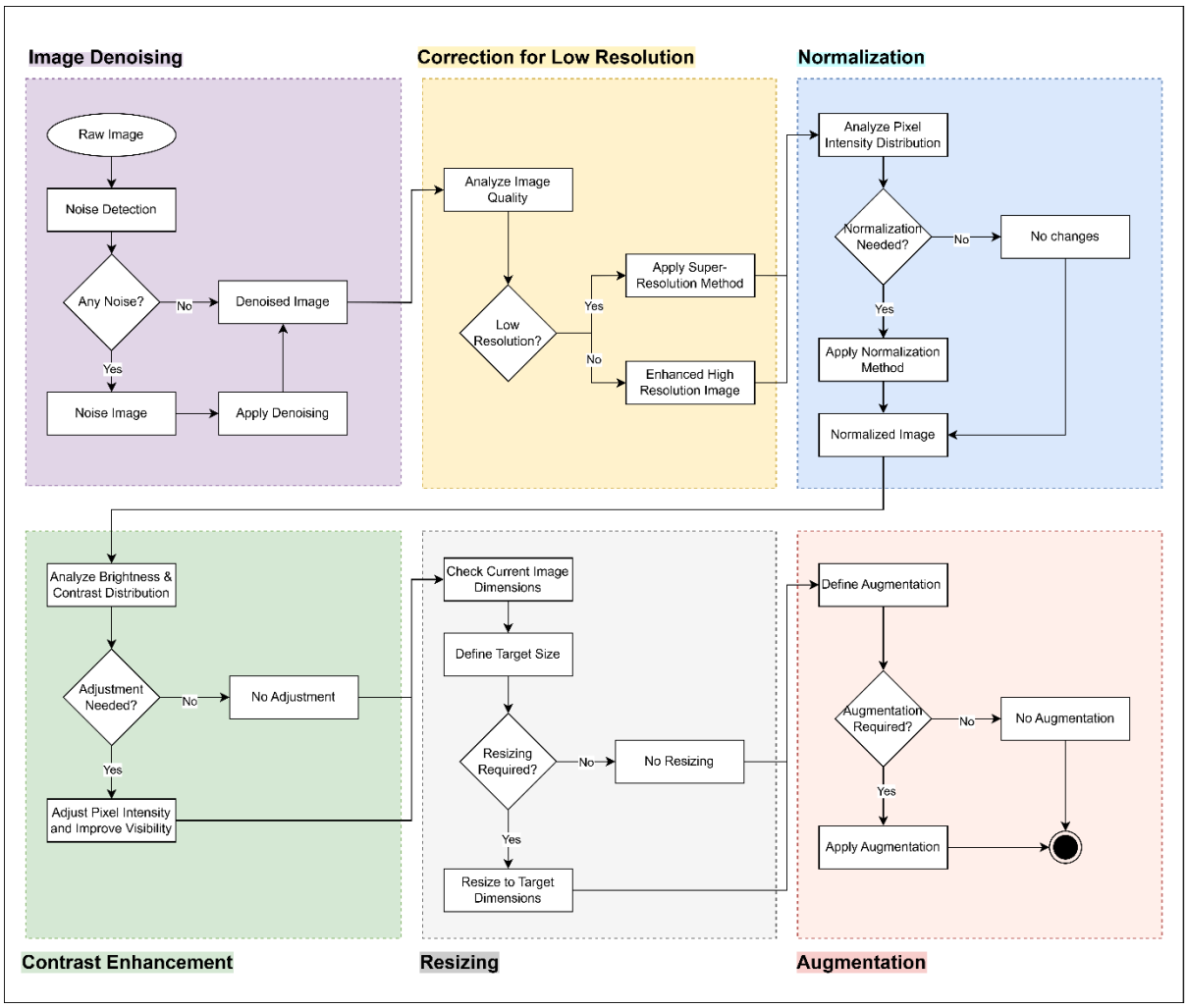
*Flow Chart for **Unstructured/ Semi-structured Textual Data** Preprocessing Techniques*



Note. Each textual data preprocessing technique solves specific textual quality issues and can combine into a workflow to improve overall image clarity and consistency. Noisy-text cleaning is the first step to remove unnecessary raw textual data such as punctuation word. Next, typos or misspelling will be address, followed by tokenization where a string of text is segment into a list of individual units. After that, lemmatization will be done by filtering the text based on worker's preference such as stop words. Finally, It will be presented using two methods which are transformer-based preprocessing and static embedding, both of which having different advantages depending on the requirements.

Figure 3

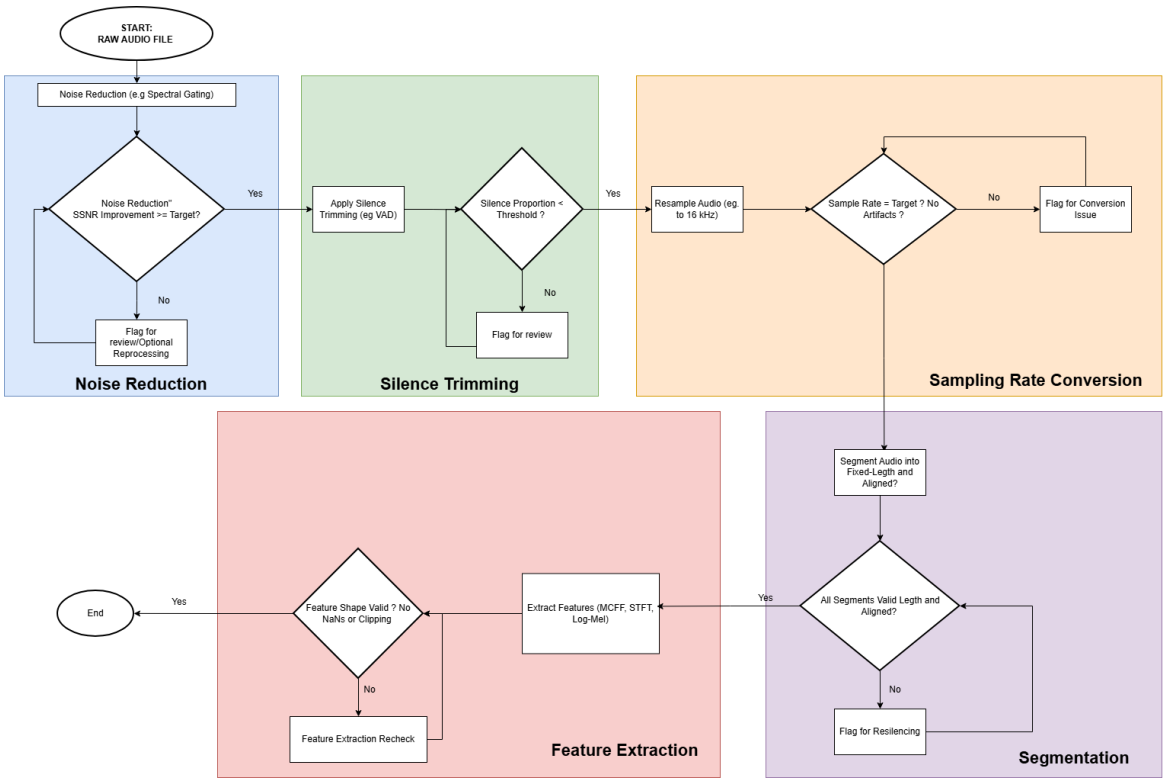
*Flow Chart for **Image Data** Preprocessing Techniques*



Note. Each image preprocessing technique solves specific image quality issues and can combine into a workflow to improve overall image clarity and consistency. The process begins with image denoising to remove unwanted noise, following by correction for low resolution to enhance image detail. After that, normalization help to standardizes pixel intensity and contrast enhancement adjusts brightness and visibility. Lastly, image resizing and augmentation is applied to produce quality image.

Figure 4

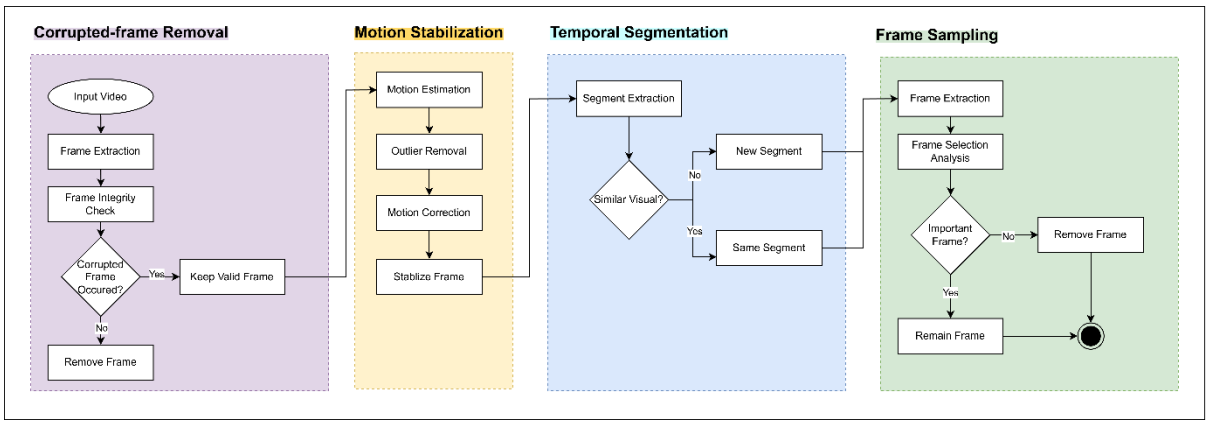
*Flow Chart for **Audio Data** Preprocessing Techniques*



Note. This flowchart provides the main steps of preprocessing of audio to machine learning pipelines. Raw audio is then reduced in noise, trimmed in silence, and resampled after which it is segmented into homogeneous frames. Characteristics like MFCC, STFT, or log-Mel spectrogram are then taken out to feed a model. SNR gain, silence thresholds, sampling conformity, segment alignment, and feature validity are checked with quality checkpoints (diamond shapes). It is a standardized audio input structure, enhanced model performance, and an abidance with current research practices (e.g., Abayomi-Alli et al., 2022; Zaman et al., 2025; Chu et al., 2023).

Figure 5

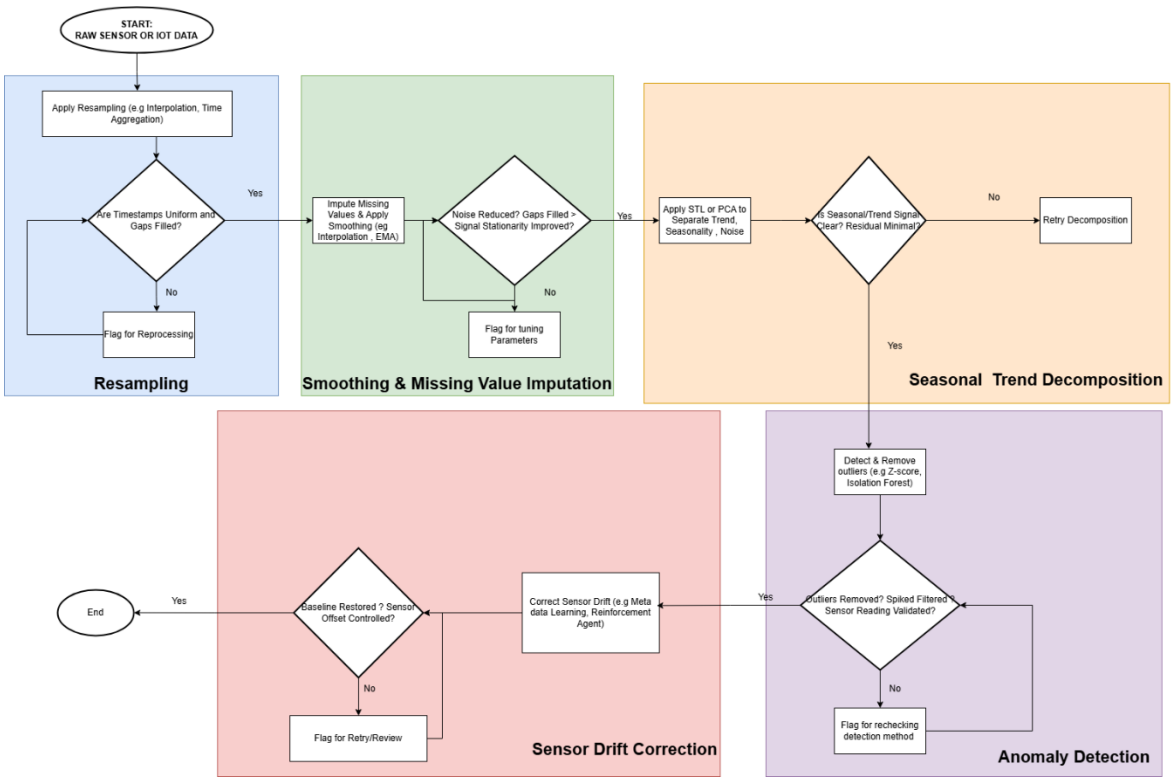
*Flow Chart for **Video Data** Preprocessing Techniques*



Note. Although each preprocessing technique solves different video data problem but can combine into a single workflow to enhance overall video quality. First, the corrupted frame will be removed. After that, motion stabilization will correct and stabilize the frames with motion issues. Other than that, temporal segmentation and frame sampling applied to preserve essential visual information while reducing redundancy.

Figure 6

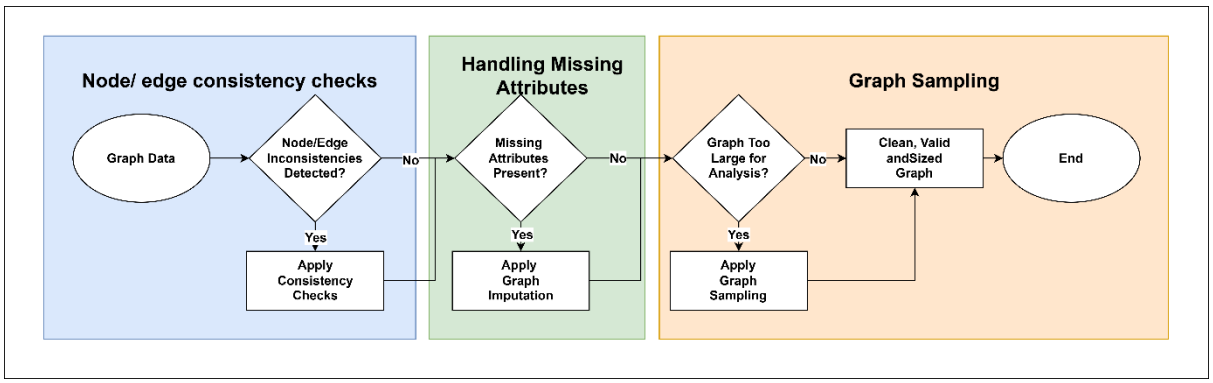
*Flow Chart for **Time-series/ Sensor/ IoT Data** Preprocessing Techniques*



Note. The flowchart is used to list the major preprocessing methods to be applied to time-series and IoT sensor data to overcome quality concerns like irregular sampling, noise, drift and signal anomalies. Resampling is initially used to normalize uneven timestamps and recreate time regularity, and then, the data is processed through the use of smoothing and imputation to heal the gaps in the data and diminish volatility. The seasonal-trend decomposition is followed to remove noise and enhance interpretability of the patterns of time variations. Anomaly detection modules eliminate spikes and incorrect measurements, and drift correction modules reestablish the long-term stability of sensors. At every stage, quality control guarantees signal integrity, fidelity, and alignment, then the cleaned data is passed by downstream modelling stages (Stefanou et al., 2025; Usmani et al., 2024; Tawakuli et al., 2025; Wang et al., 2025).

Figure 7

*Flow Chart for **Graph/Network Data** Preprocessing Techniques*



Note. Each graph data preprocessing technique solves specific image quality issues and can combine into a workflow to improve overall image clarity and consistency. The process begins by detecting inconsistent node or edge and remove unwanted noise, followed by checking missing attributes to address gaps. Next, graph sampling will be done if the graph is too large by abstracting it into smaller subgraph for analysis. Finally, the graph is clean, valid and can be prepared to use for analysis.

3.0 Comparative Analysis Table

Table 1

Comparative Analysis Table for All Preprocessing Techniques

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
		Structural Textual Data		
Missing-value Handling	Missing or incomplete entries from sensors, human error or data loss.	Restores dataset completeness, improving statistical validity.	May distort correlations if inappropriate imputation is used.	Clinical prediction models with imputed medical data (Shadbahr, 2023)
Duplicate Detection	Repeated or overlapping records caused by merging datasets or manual input	• Improves consistency • Reduces bias and data redundancy.	Can be computationally costly on large datasets.	Educational performance analytics using record linkage (Azeroual, 2022)
Scaling / Normalization	Uneven feature scales affecting model training.	Ensures equal weight for all features.	Sensitive to outliers if Min-Max is used.	SVM and neural network model performance comparison (Pinheiro, 2025)
Encoding	Non-numeric or categorical variables that models cannot process directly.	• Enables categorical data interpretation • Supports mixed data types	May increase dimensionality.	Benchmark of encoders on binary classification datasets (Matteucci., 2023)
Outlier Detection	Extreme or invalid values	Reduces influence of anomalies.	Risk of removing rare but important patterns	Anomaly detection on ML benchmark datasets (Badhan & Ganpati, 2024)
Class Imbalance Handling	Unequal class distribution leading to biased predictions	• Balances learning across classes • Improves model fairness	Oversampling may cause overfitting	Credit-risk and fraud-detection model using resampling and cost-sensitive learning (*Chen.,*

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
				2024)
Unstructured Textual Data				
Noisy-test cleaning	Irrelevant or unwanted "noise" in datasets.	Allows for the identification of meaningful patterns and insights.	Generic rules can be inflexible and may remove important domain-specific words (Kathuria et al., 2020)	• Stop word removal • Punctuation removal
Tokenization	Complex, unstructured text that is difficult to analyse.	• Creates a basic structure from cleaned text. • Makes text manageable and easier for analysis.	Overdoing tokenization can cause a significant loss of content and meaning.	Natural Language Toolkit (NLTK) (GeeksforGeeks, 2017)
Lemmatization	Large vocabulary size due to different inflectional word endings.	• Significantly decreases vocabulary volume. • Improves accurate data mining. • Reduces storage space.	Can fail if a word like new slang or a technical term is not in its dictionary. (Kathuria et al., 2020)	Stemming
Embeddings	Text is not in a numerical format that machines can understand.	• Captures the semantic meaning of words. • Enables machines to perform tasks.	A word has only one vector, so it cannot differentiate between words with multiple meanings. Ext: "bank" as in river vs. finance (Manoj Prabaharan, 2025)	Natural Language Processing (NLP)

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
Transformer-based Preprocessing	Inability to understand the nuances of language or "true meaning" of words in context.	Understands the "true meaning" by weighting the importance of all words.	Models are extremely large and require powerful hardware (Manoj Prabaharan, 2025)	Contextual encoding
		Image Data		
Denoising	Random noise caused by environmental or sensor limitations.	Removes unwanted noise and improves clarity	Over-smoothing may lead to loss of fine Details	Flow-Based Image Denoising Neural Network (Liu et al., 2022)
Contrast Enhancement	Low brightness or poor visibility	Improves image visibility, emphasizes important regions, and balances brightness.	Excessive enhancement may introduce artifacts	1. Plain 3D U-Net 2. Residual 3D U-Net 3. Pre-activation Residual 3D U-Net (Heller et al., 2021)
Resizing	Inconsistent image dimensions	Standardizes image size	Distort aspect ratio or lose fine details.	CNN-based resizer (Talebi and Milanfar, 2021)
Normalization	Uneven pixel intensity distribution	Standardizes pixel value range and improves model performance.	Reduce image contrast or dynamic range.	Deep Neural Networks (Huang et al.,2023)
Augmentation	Limited data or overfitting	Reduces overfitting	Risk of data distortion	1. Data Warping Augmentations - Geometric - Colour Space - Kernel Filters - Random Erasing

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
				- Adversarial Training - Neural Style Transfer 2. Oversampling Augmentations - Mixing Images - Feature Space Augmentation - GAN-based Augmentation (Shorten & Khoshgoftaar, 2019)
Correction of low-resolution	Blurry or low-detail images	Enhances image sharpness and restores lost details	may generate unrealistic textures	Adaptive back-projection algorithm 1 (Messina et al., 2002)
		Audio Data		
Noise Reduction	Environmental/background noise (e.g., wind, crowd, machinery) that obscures relevant audio features	Enhances clarity and intelligibility; improves classification accuracy. Essential for robust model training.	Risk of removing important sound details; real-time filtering may be computationally expensive	Speech enhancement using **VAE denoising** for voice communication *(Nogales et al., 2023; Abayomi-Alli et al., 2022)*
Silence Trimming	Long inactive or silent segments that pad or confuse time-sequenced models	Reduces irrelevant input; improves model focus and reduces training time	May mistakenly remove quiet but meaningful segments; improper trimming may cut useful signals.	Trimming dead space and normalizing amplitude before **MFCC** conversion for classification *(Chu et al., 2023; Nogales et al., 2023)*
Sampling-rate conversion	Inconsistent audio resolution across datasets that affects	Aligns time scale and resolution; ensures uniform	Poor resampling can cause aliasing or loss of	Standardizing sampling rates before **log-Mel/FFT** feature

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
	temporal alignment and feature consistency.	feature extraction across datasets; minimizes aliasing.	fidelity if rates are mismatched.	extraction in transformer audio models *(Abayomi-Alli et al., 2022; Zaman et al., 2025).*
Segmentation	Long continuous audio without labels or natural boundaries	Allows localized learning; improves recognition of short sound events; facilitates data augmentation.	Segmentation errors can cause loss of context or misalignment of sound events.	Short-frame segmentation for **CNN-based MFCC spectrogram** generation *(Chu et al., 2023)*
Feature Extraction	Raw waveform is high-dimensional and too unstructured for direct input	Preserves key spectral and temporal features; reduces dimensionality; enables accurate model learning.	noise; **STFT** is computationally heavier; requires balanced preprocessing.	**MFCC/STFT** feature extraction achieving >90% accuracy on **UrbanSound8K** with ANN (Gourisaria et al., 2024; Chu et al., 2023; Nogales et al., 2023; Zaman et al., 2025)
		Video Data		
Frame Sampling	Excess frames leading to large data size and inefficiency.	Reduces data volume and retains key visual information.	Risk of losing important temporal details if sampling rate is not optimal.	1. Interval-based 2. Pixel Intensity-based 3. Structural Feature-based 4. Semantic Feature-based 5. Shot Boundary-based (Kandhare & Gisselbrecht, 2024)
Temporal Segmentation	Continuous content that causes difficulty in scene-level analysis.	Preserves scene boundaries and improves content organization.	May misidentify transitions in videos with gradual scene changes.	Scene Separation and Data Selection algorithm - Scene Separation: based on dHash - Data Selection: Length Weighted Average

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
				Pooling (Xin et al., 2022)
Motion Stabilization	Camera shakes and unwanted motion occurred across frames.	Produces smooth and stable video sequences and improves visual quality.	Can distort objects or motion paths if stabilization is overapplied	1.0 Deep3D Stabilizer (Lee et al., 2021)
Removal of Corrupted Frames	Corrupted or unreadable frames due to transmission or storage errors.	Ensures data consistency and prevents error	Excessive removal cause discontinuity in the video sequence.	The Three-Parameter Corruption Model - P is frame corruption probability - L is corruption location - S is fragment size (Liu et al., 2023)
Time-Series/ Sensor/ IoT Data				
Resampling	Gaps in timestamps, irregular or missing sampling intervals, motion artifacts, and signal drift that distort temporal alignment.	Rebuilds consistent time intervals; stabilizes temporal resolution; allows standardized time-based analysis and synchronization of sensor streams.	May introduce interpolation bias or temporal lag if applied incorrectly on non-stationary signals.	Temporal resampling and alignment for wearable accelerometer data to correct motion irregularities *(Stefanou et al., 2025; Tawakuli et al., 2025).*
Smoothing	Noise, missing values, signal instability, and short-term fluctuations in IoT sensor readings.	Reduces random noise and outliers; stabilizes variance; enhances signal clarity for downstream forecasting.	Excessive smoothing can remove meaningful rapid changes; risk of overfitting when using complex filters.	Exponential Moving Average (EMA), spline interpolation, and Box–Cox transformations for stable sensor signals *(Usmani et al., 2024; Tawakuli et al., 2025; Wang et al., 2025)*
Seasonal-trend decomposition	Overlapping long-term trends	Separates seasonality, trend,	Decomposition accuracy	STL and FFT-based

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
	and short-term variations that obscure periodic behavior or features.	and residual components; enhances interpretability; improves model focus on meaningful patterns.	depends on data length and windowing; mis-specified trend models may distort results.	decomposition to isolate frequency-domain trends and residual noise *(Tawakuli et al., 2025)*
Anomaly Detection	Unwanted spikes, outliers, or faulty sensor readings that bias model learning.	Detects abnormal measurements; reduces false alarms; ensures model reliability and robustness.	Sensitive to threshold settings; deep models require high computational cost.	Z-score, IQR filters, isolation forest, and deep autoencoder or GAN-based anomaly detection for multivariate sensor data . *(Wang et al., 2025)*
Sensor Drift Correction	Gradual sensor drift, calibration shifts, or pipeline inefficiency affecting data reliability.	Maintains long-term stability; automates preprocessing selection; supports scalable, low-latency edge deployment.	Requires specialized frameworks; may be hardware or architecture-dependent.	Automated meta-learning preprocessing (Preptimize), edge-based modular ETL pipelines (Docker/Kubernetes), and DRL agents for real-time sensor cleaning. *(Usmani et al., 2024; Urblik et al., 2023; Mohammed et al., 2023)*
		Graph/ Network Data		
Node Consistency Checks	• Data that violates established rules. • Structural inconsistencies. • Redundancy (repeated facts or relations).	Ensures the graph respects simple integrity constraints.	Graphs are semi-structured and often do not come with a schema to specify the integrity and semantics of the data (Xue & Zou, 2022)	Dangling edge detection (GeeksforGeeks, 2024)
Handling Missing Attributes	• Gaps in a dataset. • Incompleteness of data.	• Creates a much richer and more complete dataset.	• Risk of bias from misspecified models	• Neighbour-based imputation (using

Preprocessing Technique	Targeted Data Quality Problem	Advantages	Limitations	Example Application
		• Allows data mining models to find more accurate patterns.	• Expensive computational cost (Chourib, 2025)	mean/median/mode). • GNN imputation (predicting values).
Graph Sampling	The scalability problem of large networks.	• Extracts a smaller subgraph while maintaining its key meaning. • Preserves local structures (Random Walk). • Draws representative subgraphs.	Size reduction may potentially cause losing information (Hu & Lau, 2013)	• Edge sampling • Random walk sampling

4.0 Discussion of Emerging Trends

The trend in recent years have shifted from manual tasks towards intelligent and automated systems. A major trend is the use of Generative AI for capturing complex patterns in real-world data and generating synthetic data, which is particularly useful for data augmentation. A model example is Generative Adversarial Network (GAN) to generate newer, more realistic data samples by distinguishing between real and synthetic data (Wen et al., 2024). Another significant trend is Data-Centric AI (DCAI), this framework focuses on developing, systematically engineering and maintain data quality. It ensures the accuracy, diversity and representation of data by addressing data quality issues and reduce bias, which ultimately brings better model performance and generalization (Zha et al., 2023). An example is a development team that not only tuning the model, but also systematically identifies and corrects mislabelled data in their training set or programmatically labels large unlabelled datasets to find and fix noise and bias.

5.0 Critical Evaluation & Integration

Across all data types, model performance depends on data quality. Standardisation is the first step to make inputs comparable, for example scaling tabular features, normalising image intensity, aligning audio sampling rates, resampling sensor timestamps, validating graph structure, and tokenising or lemmatising text. Noise reduction should improve clarity while preserving rare but meaningful patterns such as outliers, minority classes, and brief events. Sampling, segmentation, and resizing can reduce cost and improve coverage, yet poor settings may introduce bias or misalignment. Strong data preprocessing also involves quality checks for duplicates, corruption, anomalies, and class imbalance issues, supported by measures appropriate for each data type. Regular validation and careful documentation help to maintain consistency and avoid data drift across analysis stages. In summary, preprocessing should me modular, cautious and well-managed. Automation and augmentation may support human oversight but should not replace it, as maintaining fairness and the consistency across data sources is important for producing reliable results.

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